Topic 9 – Describing relationships

ENVX1002 Statistics in Life and Environmental Sciences

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Apr 2026

About me

- Lecturer in Agricultural Data Science
- Research topics: spatial modelling and mapping, precision agriculture, soil contamination
- How did I get here?
  - BSc (Hons) in Environmental Biology - Thesis on satellite detection of phytoplankton off Australian coast
  - PhD in Agriculture - Thesis on soil contamination mapping and spatial modelling
  - Various postdocs - Teaching, Wagyu metabolomics, soil constraints, soil carbon

Figure 1: Liana standing in a cotton field like a professional

Learning Outcomes

By the end of this course, students wil be able to:

- LO1. Implement basic reproducible research practices—including consistent data organization, documented code, and version-controlled workflows—so that their statistical analyses and results can be readily replicated and validated by others.
- LO2. Demonstrate proficiency in utilizing R and Excel to effectively explore and describe life science datasets.
- LO3. Apply parametric and non-parametric statistical inference methods to experimental and observational data using RStudio and effectively interpret and communicate the results in the context of the data.
- LO4. Put into practice both linear and non-linear models to describe relationships between variables using RStudio and Excel, demonstrating creativity in developing models that effectively represent complex data patterns.
- LO5. Articulate statistical and modelling results clearly and convincingly in both written reports and oral presentations, working effectively as an individual and collaboratively in a team, showcasing the ability to convey complex information to varied audiences.

Module overview

- **Week 9. Describing Relationships**
  - Correlation (calculation, interpretation)
  - Regression (model structure, model fitting)
  - What/when/why/how
- Week 10.

  Simple Linear Regression
  - Can we use the model? (assumptions, hypothesis testing)
  - How good is the model? (interpretation, model fit)
- Week 11.

  Multiple Linear Regression
  - Multiple Linear Regression (MLR) modelling
  - Assumptions, interpretation and the principle of parsimony
- Week 12.

Nonlinear Regression

- Common nonlinear functions
- Transformations

Module overview

- **Week 9. Describing Relationships**
  - Correlation (calculation, interpretation)
  - Regression (model structure, model fitting)
  - What/when/why/how

Example - Galton's Data

```r
library(HistData)
head(Galton)
```

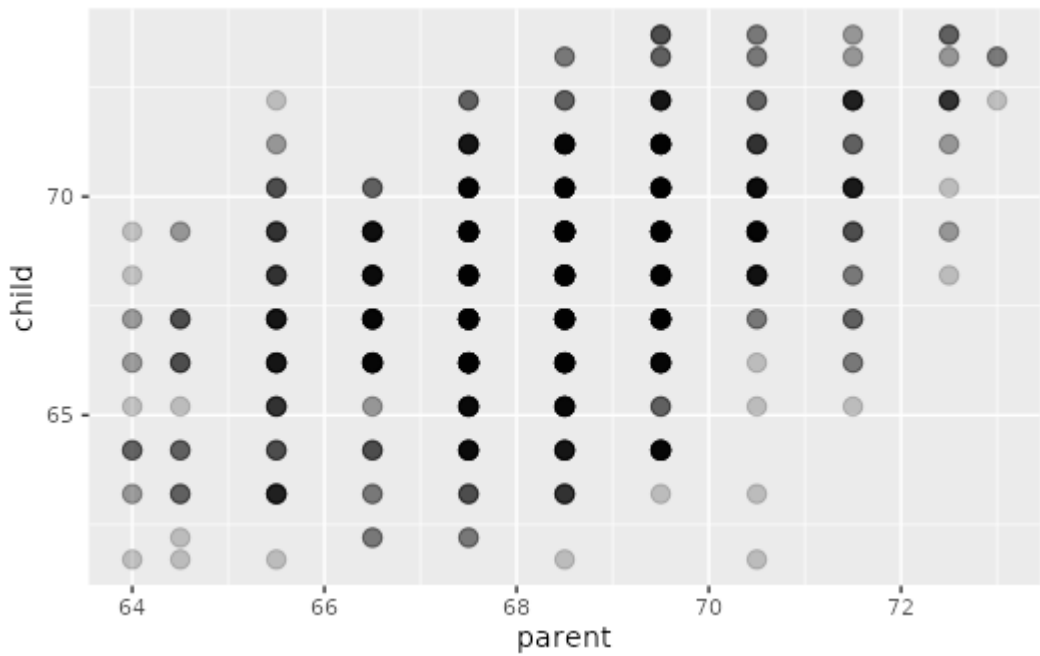
```
  parent child
1   70.5  61.7
2   68.5  61.7
3   65.5  61.7
4   64.5  61.7
5   64.0  61.7
6   67.5  62.2
```

- 928 children of 205 pairs of parents
- Average height of both parents and child measured in inches
- Size classes were binned (hence data looks discrete)

Example - Galton's Data

We can visually inspect the relationship between the two variables using a scatterplot:

```
library(ggplot2)
ggplot(Galton, aes(x = parent, y = child)) +
    geom_point(alpha = .2, size = 3)
```



... but this is not a very good way to assess the strength of the relationship between the two variables.

Is the relationship:

- Linear?
- Positive or negative?
- Weak, moderate or strong?

Correlation

Correlation Coefficient

The **correlation coefficient** is a number between −1 and 1 that describes the relationship between two **continuous** variables.

- **Direction**:
  - Positive – both variables increase together
  - Negative – one variable increases as the other decreases
- **Strength**:
  - $-1 \longrightarrow$ perfect negative relationship
  - $0 \longrightarrow$ no relationship
  - $1 \longrightarrow$ perfect positive relationship
  - Common terms: *weak* (~0.1–0.3), *moderate* (~0.4–0.6) or *strong* (~0.7–1.0) – useful but subjective!

Pearson's Correlation Coefficient

- Karl Pearson developed the **correlation coefficient** in 1800s (based on the work by Francis Galton)

Pearson's Correlation Coefficient

Pearson's Correlation (r) Formula:

$$r = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2 \sum_{i=1}^{n}(y_i - \bar{y})^2}}$$

Covariance divided by the product of the standard deviations.

Pearson's r is:

- The most commonly used correlation coefficient
- For normally distributed data only (parametric)
- For linear relationships only (i.e. a straight line)

Luckily we have Excel and R

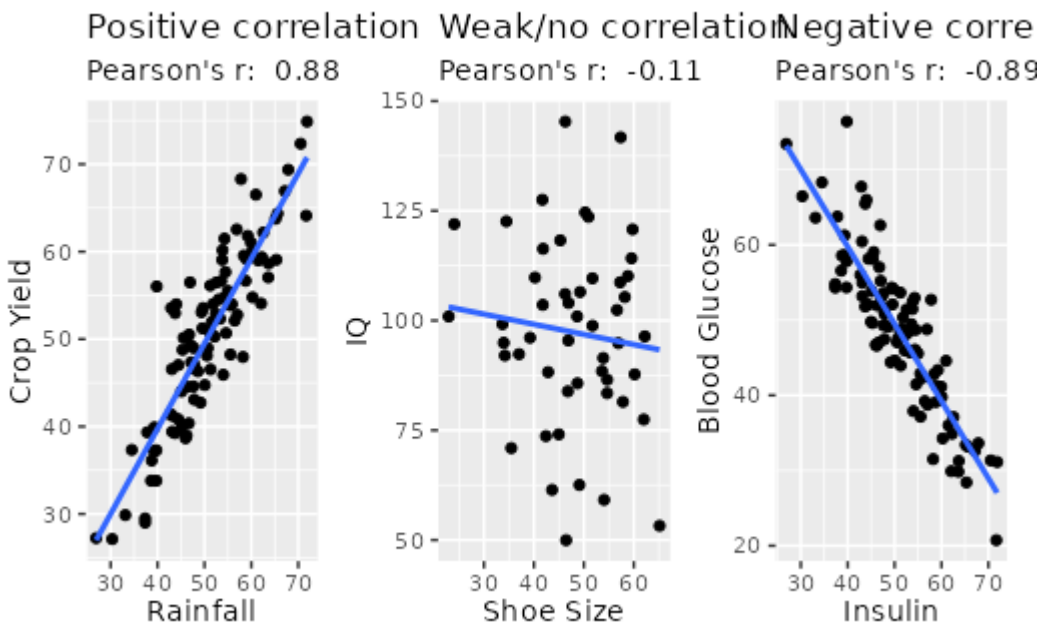
- Excel: `=CORREL()` formula, or use the Analysis Toolpak
- R: `cor()` function
  - By default this calculates Pearson's correlation coefficient
  - Can also calculate other types of correlation (`cor(x, y, method = "spearman")`)

Scatterplots give us a visual understanding

```r
# Generate synthetic data
set.seed(123)
rainfall <- rnorm(100, mean = 50, sd = 10)
plant_growth <- rainfall + rnorm(100, mean = 0, sd = 5)

# Calculate correlation coefficient
correlation_coef <- cor(rainfall, plant_growth)

# Plot data using ggplot2
library(ggplot2)
p1 <- ggplot(data = data.frame(rainfall, plant_growth), aes(x = rainfall, y = plant_growth)) +
    geom_point() +
    geom_smooth(method = "lm", se = FALSE) +
    labs(
        title = "Positive correlation",
        x = "Rainfall", y = "Crop Yield",
        subtitle = paste("Pearson's r: ", round(correlation_coef, 2))
    )
```

```r
# Generate synthetic data
set.seed(123)
insulin <- rnorm(100, mean = 50, sd = 10)
blood_glucose <- 100 - insulin + rnorm(100, mean = 0, sd = 5)

# Calculate correlation coefficient
correlation_coef <- cor(insulin, blood_glucose)

# Plot data using ggplot2
p2 <- ggplot(data = data.frame(insulin, blood_glucose), aes(x = insulin, y = blood_glucose)) +
    geom_point() +
    geom_smooth(method = "lm", se = FALSE) +
    labs(
        title = "Negative correlation",
        x = "Insulin", y = "Blood Glucose",
        subtitle = paste("Pearson's r: ", round(correlation_coef, 2))
    )
library(patchwork)

# Set the seed for reproducibility
set.seed(1249)
```

```r
# Generate the data
x <- rnorm(50, mean = 50, sd = 10)
y <- rnorm(50, mean = 100, sd = 20)
df1 <- data.frame(x, y) # combine the two variables into a dataframe
correlation_coef <- cor(x, y) # calculate the correlation coefficient

# Scatter plot
p3 <- ggplot(df1, aes(x, y)) +
    geom_point() +
    geom_smooth(method = "lm", se = FALSE) +
    labs(
        title = "Weak/no correlation",
        x = "Shoe Size", y = "IQ",
        subtitle = paste("Pearson's r: ", round(correlation_coef, 2))
    )

p1 + p3 + p2
```



(Note: not real data)

A **weak** or **nonexistent** relationship is one where the correlation coefficient is close to 0.

A **moderate** relationship falls closer to 0.5.

A **strong** relationship is one where the correlation coefficient is close to −1 or 1.

The exact values are subjective and vary between fields of study.

Example - Galton's original plots

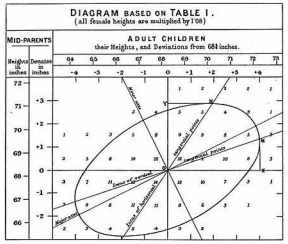
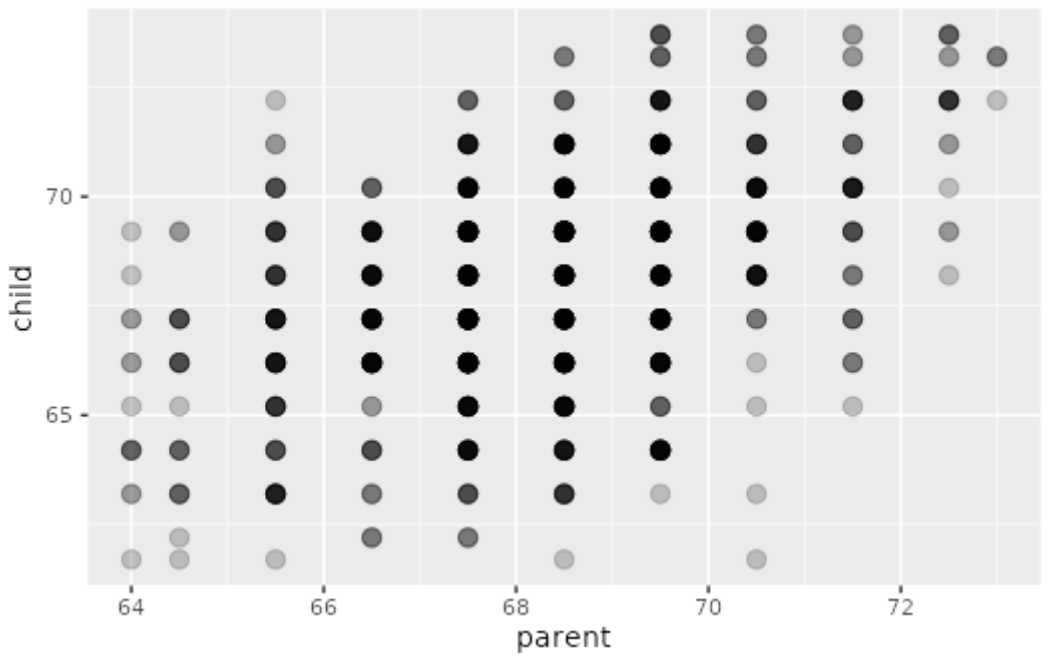


Figure 1: Galton's Results

Example - Galton's data in R

```r
library(ggplot2)

ggplot(Galton, aes(x = parent, y = child)) +
  geom_point(alpha = .2, size = 3)
cor(Galton$parent, Galton$child)
```




```
[1] 0.4587624
```

Is the relationship:

- Linear? **Seems to be**
- Positive or negative? **Positive**
- Weak, moderate or strong? **Moderate**

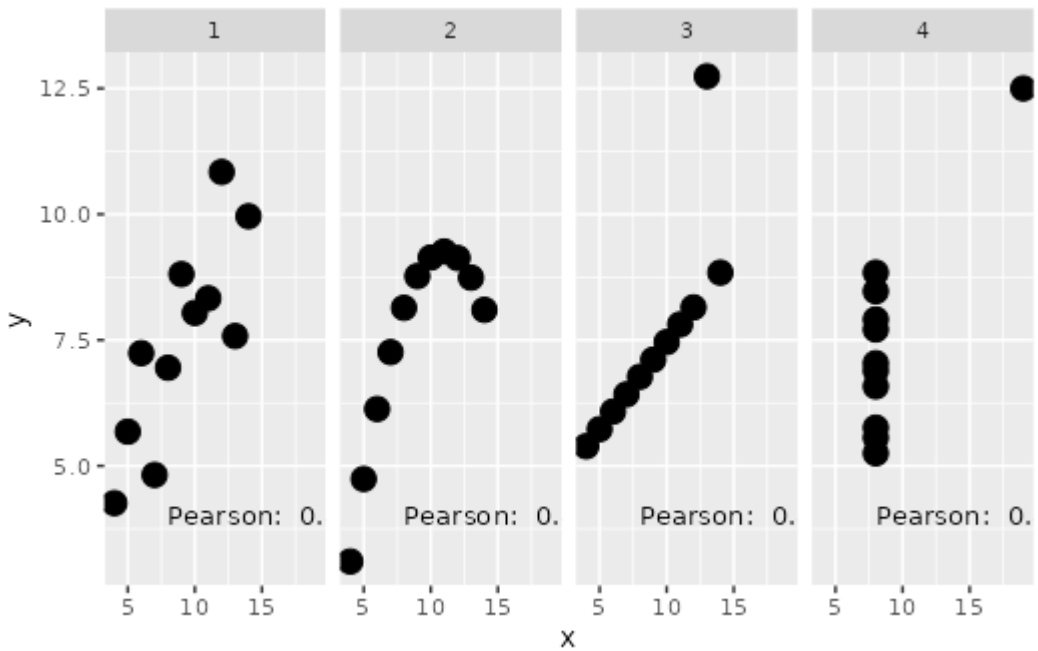
In words: “There is a moderate positive linear relationship between the height of parents and the height of their children.”

Anscombe's Quartet

A set of four datasets that are nearly identical in simple descriptive statistics but have very different distributions.

```
library(tidyverse)

# Compute Pearson, Spearman, and Kendall correlations for each set
cor_values <- anscombe %>%
  pivot_longer(everything(),
               names_to = c(".value", "set"),
               names_pattern = "(.)(.)") %>%
  group_by(set) %>%
  summarise(
    pearson = cor(x, y, method = "pearson"))

# Create the plot with the correlation values added as text annotations
anscombe %>%
  pivot_longer(everything(),
               names_to = c(".value", "set"),
               names_pattern = "(.)(.)") %>%
  ggplot(aes(x = x, y = y)) +
```

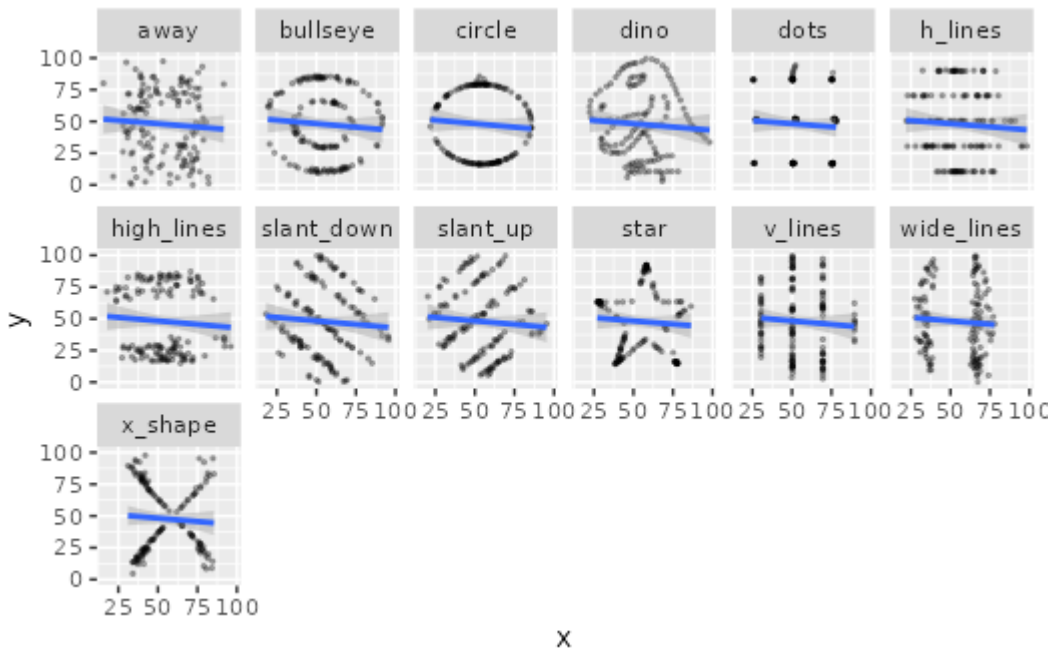
```r
  geom_point(size = 4) +
  facet_wrap(~set, ncol = 4) +
  # Add text annotations for Pearson, Spearman, and Kendall correlations
  geom_text(data = cor_values, aes(x = 8, y = 4, label =
    paste("Pearson: ", round(pearson, 2))),
    inherit.aes = FALSE, size = 3.5, color = "black", hjust = 0)
```



Correlation coefficients are **not reliable** in inferring the 'type' of relationship between variables – we must visualise.

Datasaurus Dozen

```r
library(datasauRus)
ggplot(datasaurus_dozen, aes(x = x, y = y)) +
    geom_point(size = .5, alpha = .3) +
    geom_smooth(method = "lm", se = TRUE) +
    facet_wrap(~dataset, ncol = 6)
```



All of these data have a correlation coefficient close to zero!

Monotonic vs linear relationships

- **Monotonic**: a relationship that is consistently increasing or decreasing
- **Linear**: a relationship that is increasing or decreasing at a *constant rate* i.e. a straight line

Pearson's correlation is only for linear relationships.

Spearman's rank and **Kendall's tau** are correlation coefficients for all monotonic relationships.

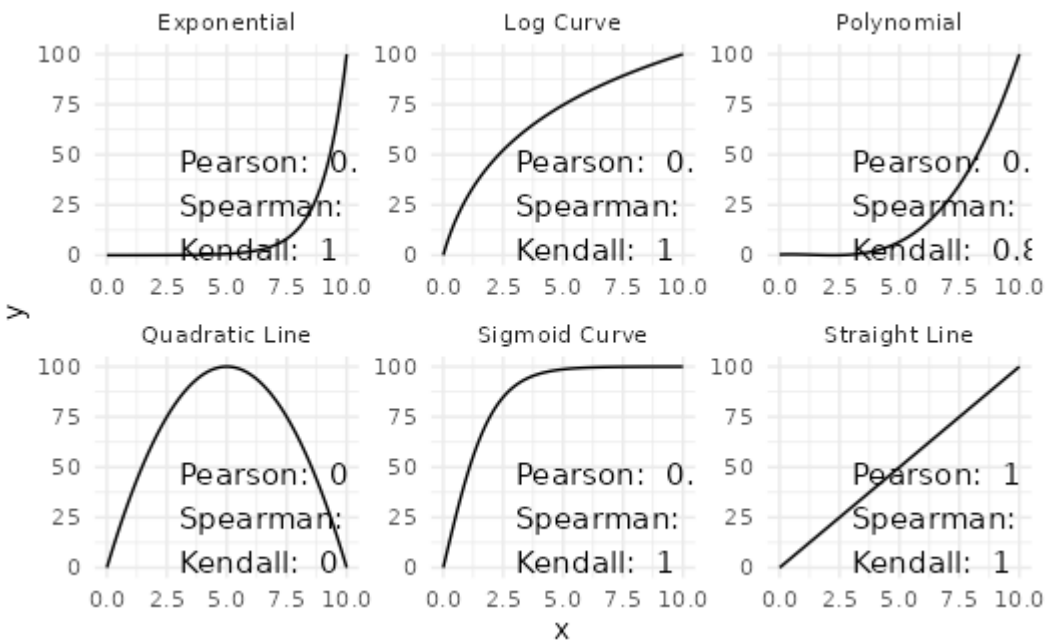
- Works with non-parametric data
- More 'conservative' i.e. values can be smaller in magnitude, but more robust against outliers

```r
library(tidyverse)

set.seed(123)
x <- seq(0, 10, length.out = 100)
x_length <- length(x)

# Create the data for each relationship
y_straight <- 2 * x + 5
y_exponential <- exp(x)
y_log <- logb(x+1, base = 2)
```

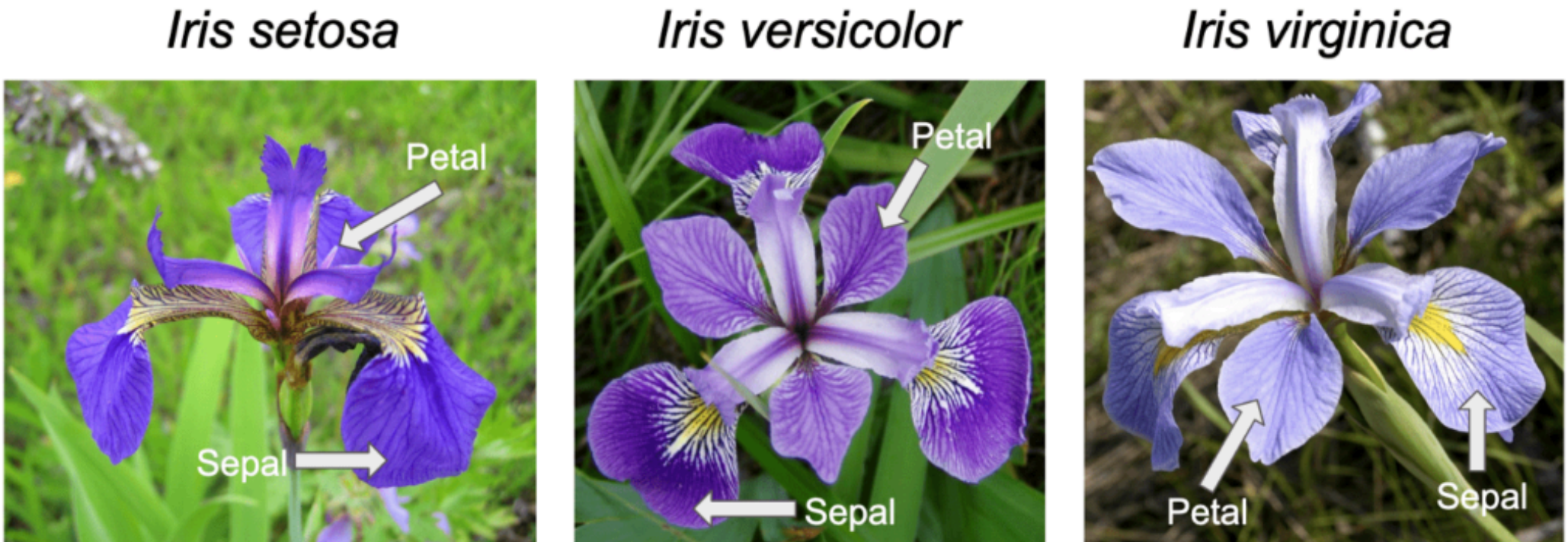
```r
y_sigmoid <- 1 / (1 + exp(-x))
y_quadratic <- -1 * (x - 5)^2 + 100
y_polynomial <- x^3 - 4 * x^2 + 3 * x + 2

# Function to scale y-values to a specific range [0, 100]
scale_to_range <- function(y_values, new_min = 0, new_max = 100) {
  old_min <- min(y_values)
  old_max <- max(y_values)
  scaled_values <- (y_values - old_min) / (old_max - old_min) * (new_max - new_min) + new_min
  return(scaled_values)
}

# Apply scaling to each relationship
y_straight_scaled <- scale_to_range(y_straight)
y_exponential_scaled <- scale_to_range(y_exponential)
y_log_scaled <- scale_to_range(y_log)
y_sigmoid_scaled <- scale_to_range(y_sigmoid)
y_quadratic_scaled <- scale_to_range(y_quadratic)
y_polynomial_scaled <- scale_to_range(y_polynomial)

# Combine the data into a tibble
```

```r
data <- tibble(
  x = rep(x, 6),
  y = c(y_straight_scaled, y_exponential_scaled, y_log_scaled, y_sigmoid_scaled,
y_quadratic_scaled, y_polynomial_scaled),
  type = rep(c("Straight Line", "Exponential", "Log Curve", "Sigmoid Curve", "Quadratic Line",
"Polynomial"), each = x_length)
)

# Calculate the correlation coefficients for each relationship type
cor_values <- data %>%
  group_by(type) %>%
  summarise(
    pearson = cor(x, y, method = "pearson"),
    spearman = cor(x, y, method = "spearman"),
    kendall = cor(x, y, method = "kendall")
  )

# Plot the data and add correlation coefficients
ggplot(data, aes(x = x, y = y)) +
  geom_line() +
  facet_wrap(~type, scales = 'free') +
```

```r
  geom_text(data = cor_values, aes(x = 3, y =25, label =
                                   paste("Pearson: ", round(pearson, 2),
                                         "\nSpearman: ", round(spearman, 2),
                                         "\nKendall: ", round(kendall, 2))),
            inherit.aes = FALSE, size = 4, color = "black", hjust = 0) +
  theme_minimal()
```



Example - `iris` Dataset



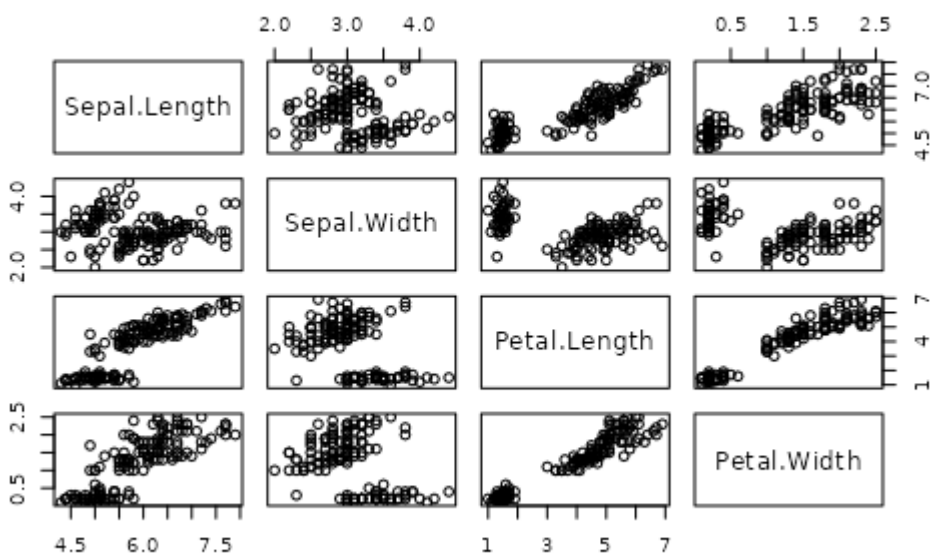

```r
head(iris)
```

```
  Sepal.Length Sepal.Width Petal.Length Petal.Width Species
1          5.1         3.5          1.4         0.2  setosa
2          4.9         3.0          1.4         0.2  setosa
3          4.7         3.2          1.3         0.2  setosa
4          4.6         3.1          1.5         0.2  setosa
5          5.0         3.6          1.4         0.2  setosa
6          5.4         3.9          1.7         0.4  setosa
```

Example - `iris` Dataset

Which correlation coefficient do we use?

```
plot(iris[,-5])
```



Example - `iris` Dataset

```r
cor(iris[,-5]) |> round(2)
```

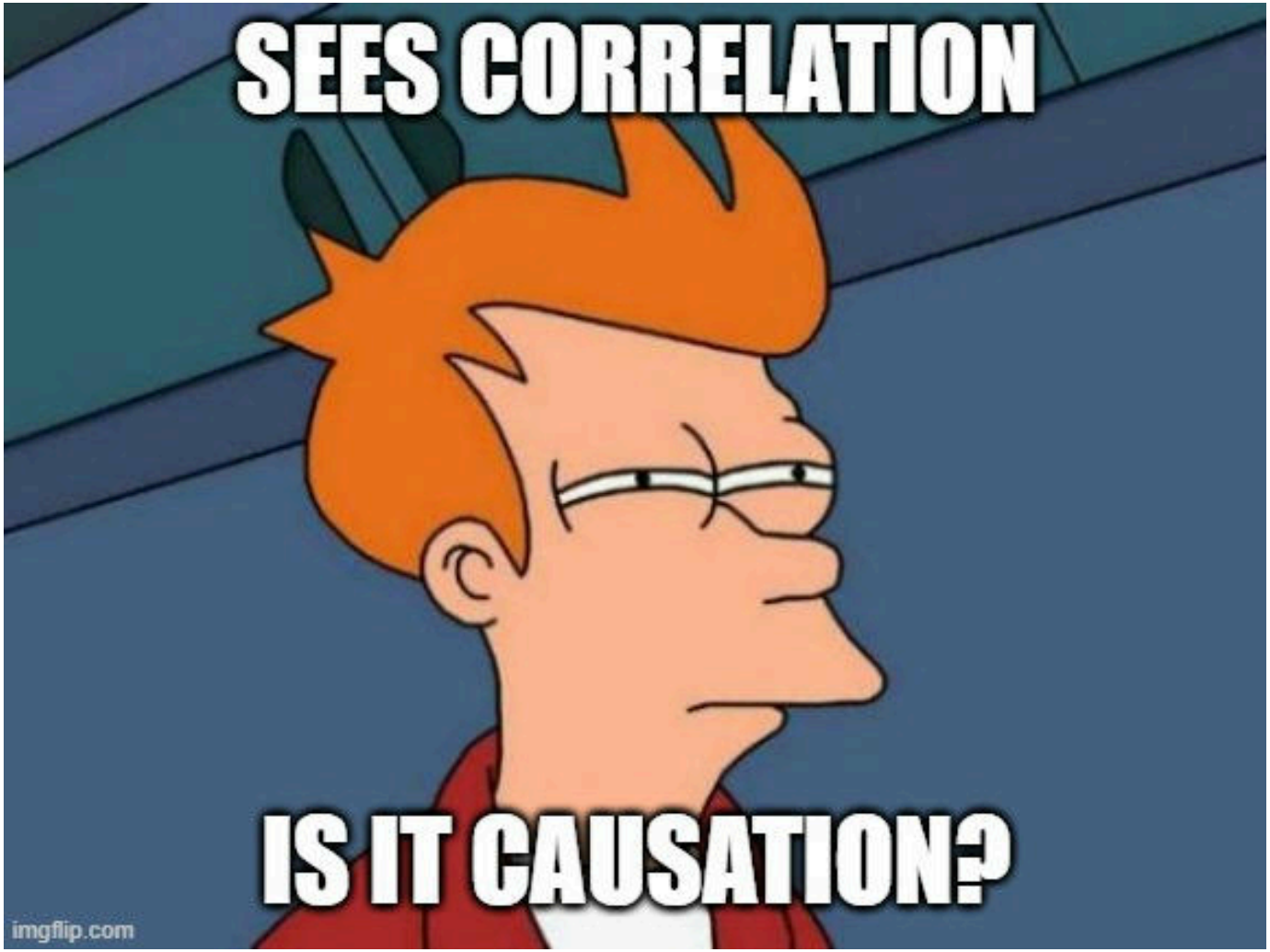
```
             Sepal.Length Sepal.Width Petal.Length Petal.Width
Sepal.Length         1.00       -0.12         0.87        0.82
Sepal.Width         -0.12        1.00        -0.43       -0.37
Petal.Length         0.87       -0.43         1.00        0.96
Petal.Width          0.82       -0.37         0.96        1.00
```

Correlation analysis **identifies** variables that have a strong linear relationship **quickly and easily**.

- We want to *predict* the `Petal.Width` of an iris flower
- Which variables would be *good predictors*?
  - Either `Petal.Length` and `Sepal Length` would be good predictors
- Which variables are likely to present issues for *model fitting*?
  - e.g. `Petal.Length` and `Sepal Length` are strongly correlated - including both would cause multicollinearity – (L11 MLR)

Correlation $\neq$ causation

Spurious correlations: a relationship between two variables does not imply that one causes the other.


SEES CORRELATION
IS IT CAUSATION?
imgflip.com

What comes after correlation?

- We have data with two or more numerical variables
- We conduct correlation analysis to describe *possible* linear relationships
  - Fast, easy, interpretable, and widely used

But, we can't infer causation: - Is there *reason() to expect a relationship between the two variables? - Do you have a* hypothesis* about the relationship between the two variables?

If we have a hypothesis about the relationship between two variables, we can use **regression analysis** to test it.

Regression modelling

- Regression is a *statistical* method to fit a model to data
- In linear regression this is a straight line that best fits the data – i.e. line of best fit

Why regression?

Describe the relationship between two variables

What is the relationship between a response variable Y and a predictor variable x?

Common Terms:

Y = response, independent variable, target, outcome etc.

x = predictor, dependent variable, feature, input etc.

Explain the relationship between two variables

How much variation in Y can be explained by a relationship with x?

Predict the value of a response variable

What is the value of Y for a given value of x?

Often we can easily measure/obtain x but not Y, so we need to *predict* Y from x.

A gateway to the world of modelling

Many types of regression models exist:

- **Simple linear regression (one predictor i.e. x)**
- Multiple linear regression (more than one predictor)
- Non-linear regression, using functions such as polynomials, exponentials, logarithms, etc.

...

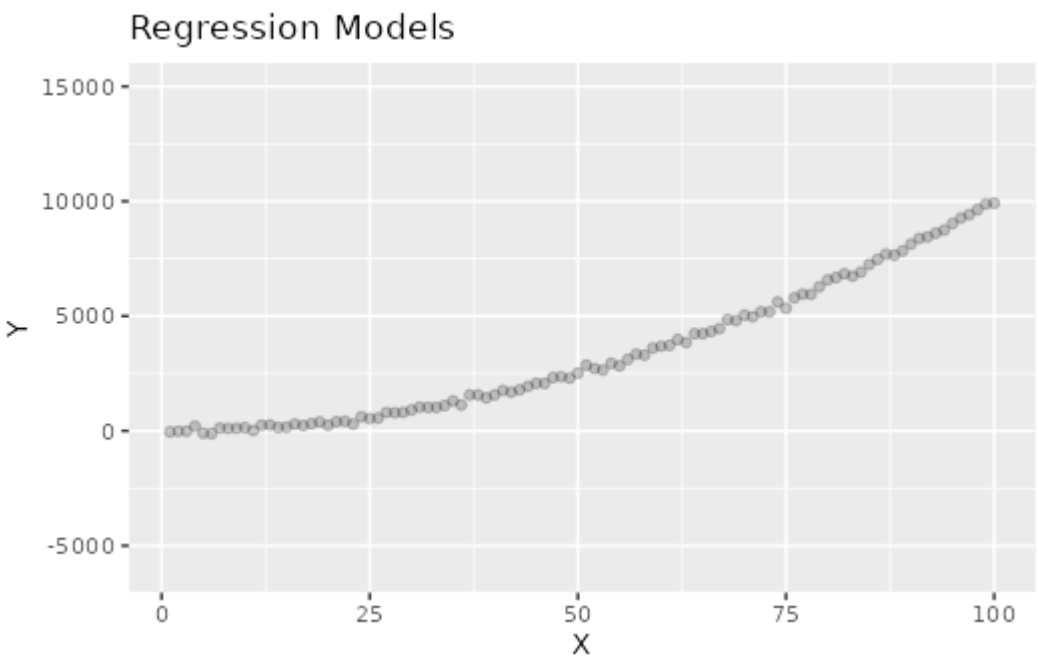
Asking ChatGPT for help with the next slide:

> Using R code, can you generate some data that is useful to demonstrate simple linear regression, multiple linear regression, polynomial, exponential and logarithmic regressions in ggplot2?

> Sure! Here's an example code that generates a sample dataset and visualizes it using ggplot2 library in R.

Visualising regression models

```r
library(ggplot2)

# Generate sample data
set.seed(136)
x <- 1:100
y <- x^2 + rnorm(100, sd = 100)

# Plot the data and regression lines
ggplot(data.frame(x = x, y = y), aes(x, y)) +
    ggtitle("Regression Models") +
    geom_point(alpha = .2) +
    xlab("X") +
    ylab("Y") +
    ylim(-6000, 15000)
```


Regression Models
15000
10000
5000
0
-5000
Y
0
25
50
75
100
X

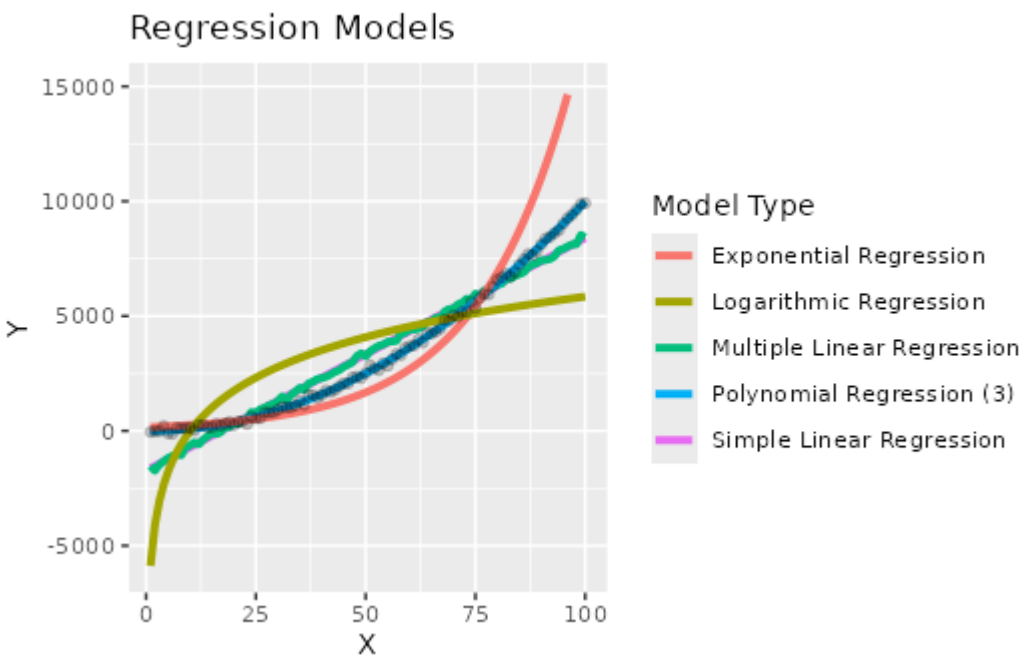
Visualising regression models

```r
library(ggplot2)

# Generate sample data
set.seed(136)
x <- 1:100
y <- x^2 + rnorm(100, sd = 100)

# Define regression functions
slr <- function(x, y) {
    mod <- lm(y ~ x)
    return(list(
        data.frame(x = x, y = predict(mod), model_type = "Simple Linear Regression"),
        paste("y =", round(coefficients(mod)[[2]], 2), "x +", round(coefficients(mod)[[1]], 2))
    ))
}

mlr <- function(x, y, z) {
    mod <- lm(y ~ x + z)
    return(list(
```

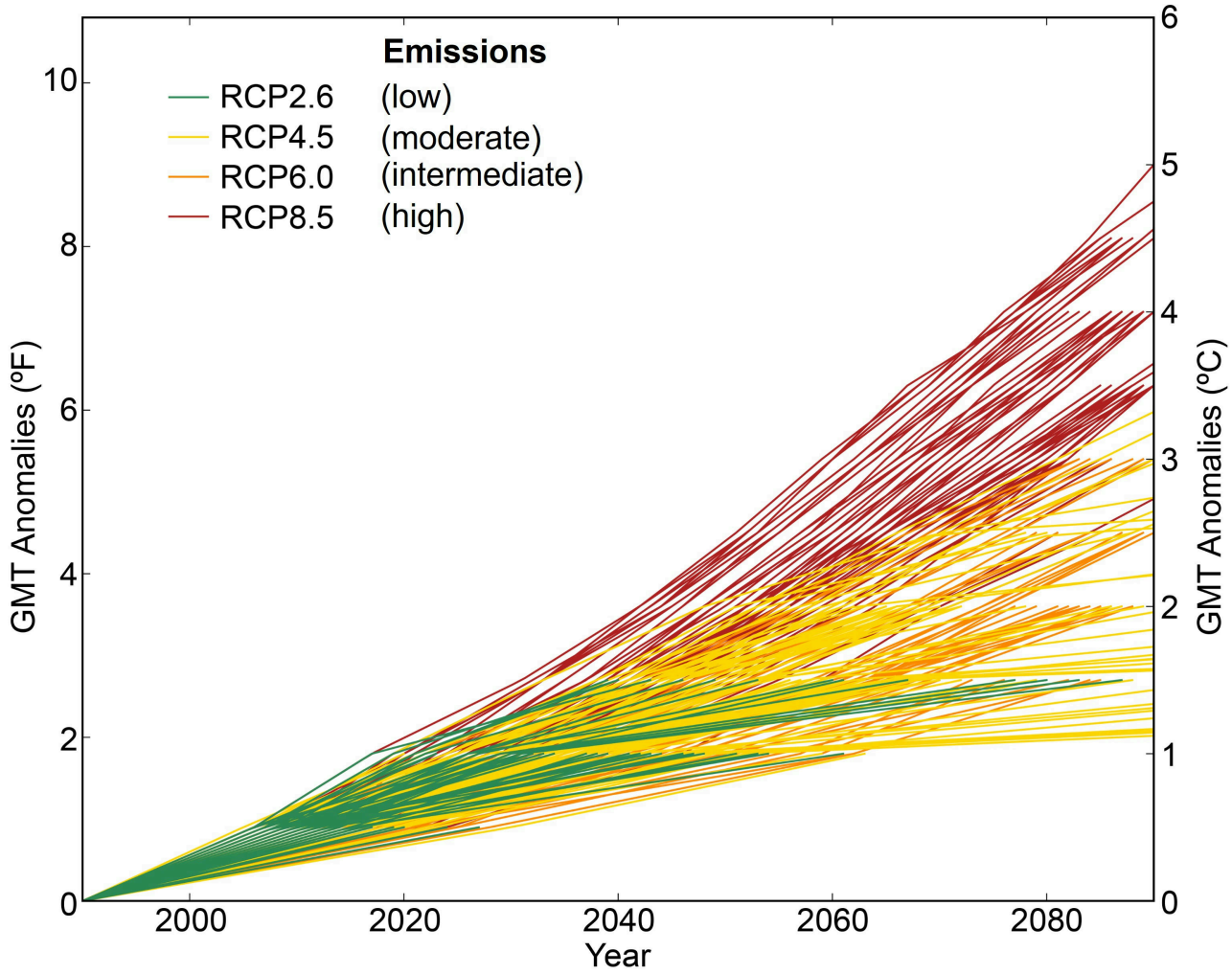
```r
        data.frame(x = x, z = z, y = predict(mod), model_type = "Multiple Linear Regression"),
        paste("y =", round(coefficients(mod)[[3]], 2), "x +", round(coefficients(mod)[[2]], 2), "z
+", round(coefficients(mod)[[1]], 2))
    ))
}

poly_reg <- function(x, y, degree) {
    mod <- lm(y ~ poly(x, degree, raw = TRUE))
    x_new <- seq(min(x), max(x), length.out = 100)
    y_new <- predict(mod, newdata = data.frame(x = x_new))
    return(list(
        data.frame(x = x_new, y = y_new, model_type = paste("Polynomial Regression (", degree, ")",
sep = "")),
        paste(paste("x^", degree, sep = ""), ":", paste(round(coefficients(mod), 2), collapse = " +
"))
    ))
}

exp_reg <- function(x, y) {
    mod <- lm(log(y) ~ x)
    x_new <- seq(min(x), max(x), length.out = 100)
```

```r
    y_new <- exp(predict(mod, newdata = data.frame(x = x_new)))
    return(list(
        data.frame(x = x_new, y = y_new, model_type = "Exponential Regression"),
        paste("y =", round(exp(coefficients(mod)[[2]]), 2), "* e^(", round(coefficients(mod)[[1]],
2), "x", ")")
    ))
}

log_reg <- function(x, y) {
    mod <- lm(y ~ log(x))
    x_new <- seq(min(x), max(x), length.out = 100)
    y_new <- predict(mod, newdata = data.frame(x = x_new))
    return(list(
        data.frame(x = x_new, y = y_new, model_type = "Logarithmic Regression"),
        paste("y =", round(coefficients(mod)[[2]], 2), "* log(x) +", round(coefficients(mod)[[1]],
2))
    ))
}

# Create regression line dataframes and equations
reg_data <- list(slr(x, y), mlr(x, y, rnorm(100, sd = 10)), poly_reg(x, y, 3), exp_reg(x, y),
```

```r
log_reg(x, y))
reg_eqs <- sapply(reg_data, function(x) x[[2]])

# Plot the data and regression lines
ggplot(data.frame(x = x, y = y), aes(x, y)) +
    lapply(seq_along(reg_data), function(i) geom_line(data = reg_data[[i]][[1]], aes(x, y, color =
reg_data[[i]][[1]]$model_type), linewidth = 1.4)) +
    ggtitle("Regression Models") +
    geom_point(alpha = .2) +
    xlab("X") +
    ylab("Y") +
    scale_color_discrete(name = "Model Type") +   # This line adds the title to the legend
    ylim(-6000, 15000) +
    theme(legend.position = "right")  # Modify this to display the legend (default position is
'right')
```

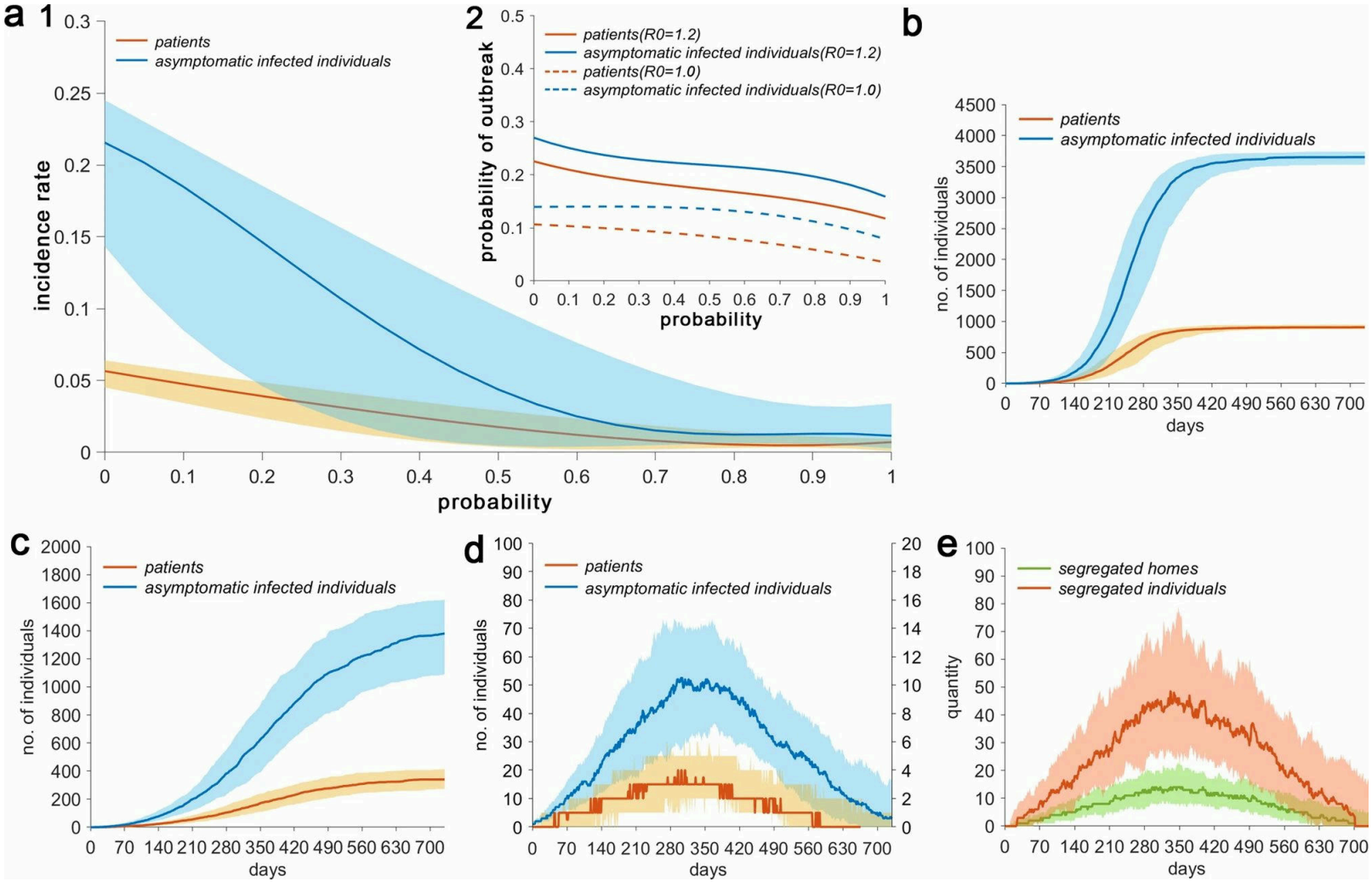

Regression Models
15000
10000
5000
0
-5000
Y
0
25
50
75
100
X
Model Type
Exponential Regression
Logarithmic Regression
Multiple Linear Regression
Polynomial Regression (3)
Simple Linear Regression

Example: climate change modelling


Emissions
RCP2.6 (low)
RCP4.5 (moderate)
RCP6.0 (intermediate)
RCP8.5 (high)
GMT Anomalies (ºF)
GMT Anomalies (ºC)
Year
0
2
4
6
8
10
0
1
2
3
4
5
6
2000
2020
2040
2060
2080

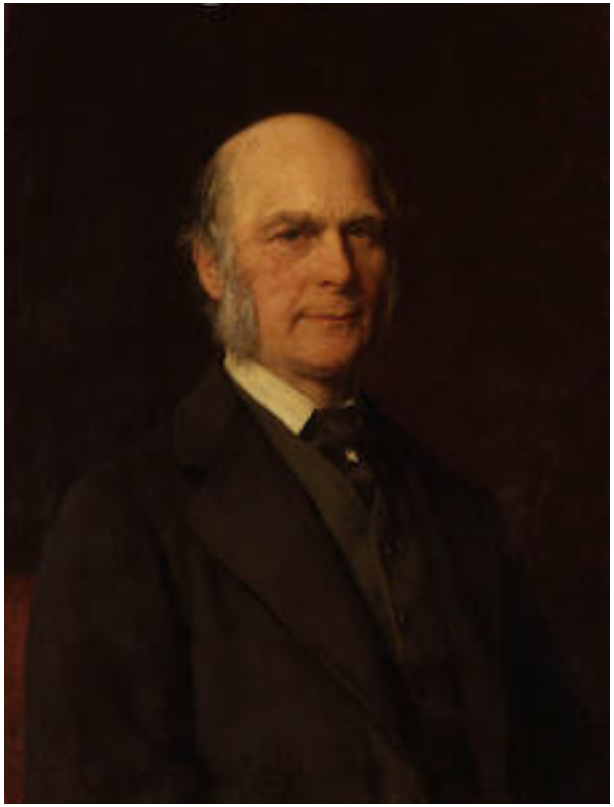
Source: https://science2017.globalchange.gov/chapter/4/

Example: COVID-19 transmission modelling


a
1
patients
asymptomatic infected individuals
incidence rate
probability
2
patients(R0=1.2)
asymptomatic infected individuals(R0=1.2)
patients(R0=1.0)
asymptomatic infected individuals(R0=1.0)
probability of outbreak
probability
b
patients
asymptomatic infected individuals
no. of individuals
days
c
patients
asymptomatic infected individuals
no. of individuals
days
d
patients
asymptomatic infected individuals
no. of individuals
days
e
segregated homes
segregated individuals
quantity
days

Source: https://www.nature.com/articles/s41598-021-84893-4/figures/1

How does regression work?



Adrien-Marie Legendre, Carl Friedrich Gauss, Francis Galton

Note

Many other people contributed to the development of regression analysis, but these three are the most well-known.

How does regression work?

- **Method of least squares** first theorised by Adrien-Marie Legendre in 1805
- **Technique of least squares** first used by Carl Friedrich Gauss in 1809 (to fit a parabola to the orbit of the asteroid Ceres)
- **Model fitting** first published by Francis Galton in 1886 (predicting the height of a child from the height of the parents)

Least squares

When we fit a line, there is error between the observed and predicted values.

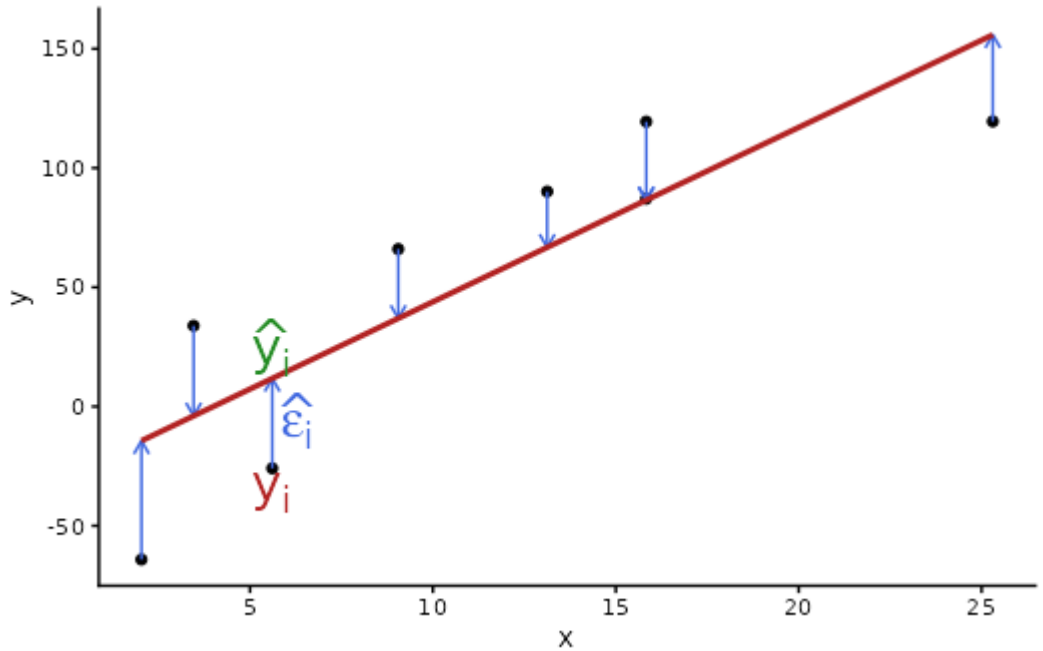
$$\hat{\epsilon_i} = y_i - \hat{y}_i$$

The **method of least squares** fits a line of best fit by minimising the sum of the squared errors.

$$\sum_{i=1}^{n} (\hat{\epsilon_i})^2$$

```r
# simulate example data

set.seed(340)
x <- runif(8, 0, 30)
y <- 5 * x + rnorm(8, 0, 40)
df <- data.frame(x, y)

# fit linear model, add residual vertical lines as arrows
mod <- lm(y ~ x, data = df)
p1 <- ggplot(df, aes(x, y)) +
  geom_point() +
  geom_segment(aes(xend = x, yend = fitted(mod)),
    arrow = arrow(length = unit(0.2, "cm")),
    color = "royalblue"
  ) +
  labs(x = "x", y = "y")

p1 +
  geom_smooth(method = "lm", se = FALSE, color =
```

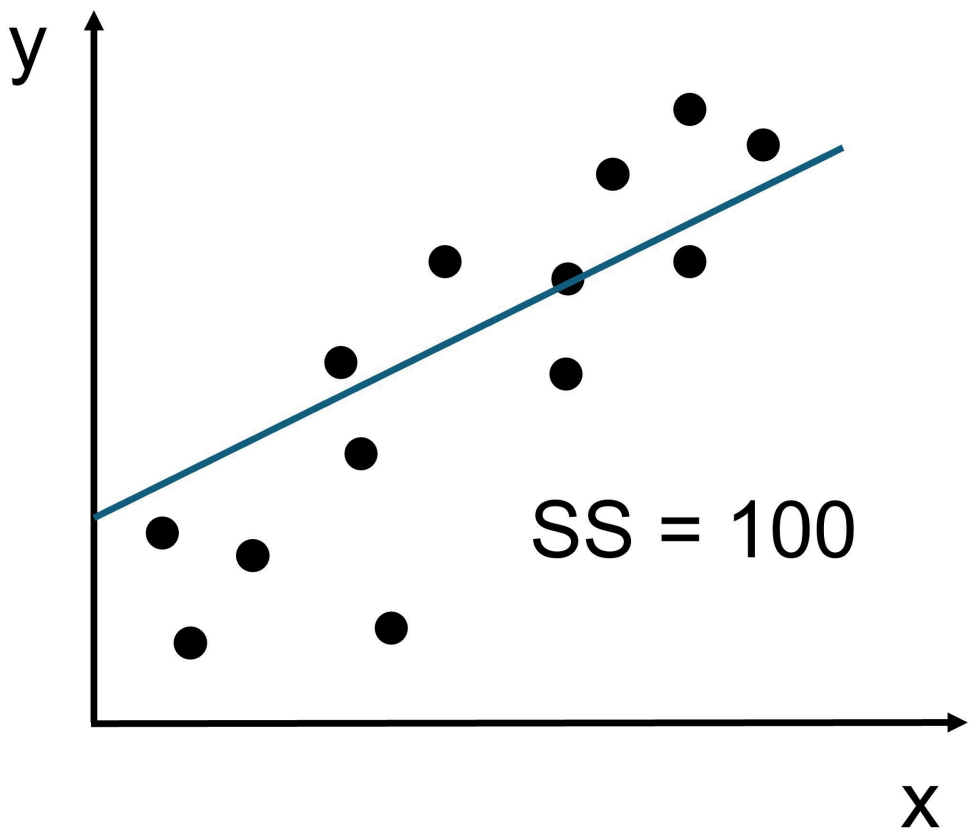
```
"firebrick") +
  annotate("text",
    x = 6.3, y = -6, size = 7,
    label = expression(hat(epsilon[i])), colour =
"royalblue"
  ) +
  annotate("text",
    x = 5.6, y = 25, size = 7,
    label = expression(hat(y[i])), colour = "forestgreen"
  ) +
  annotate("text",
    x = 5.6, y = -36, size = 7,
    label = expression(y[i]), colour = "firebrick"
  ) +
  theme_classic()
```

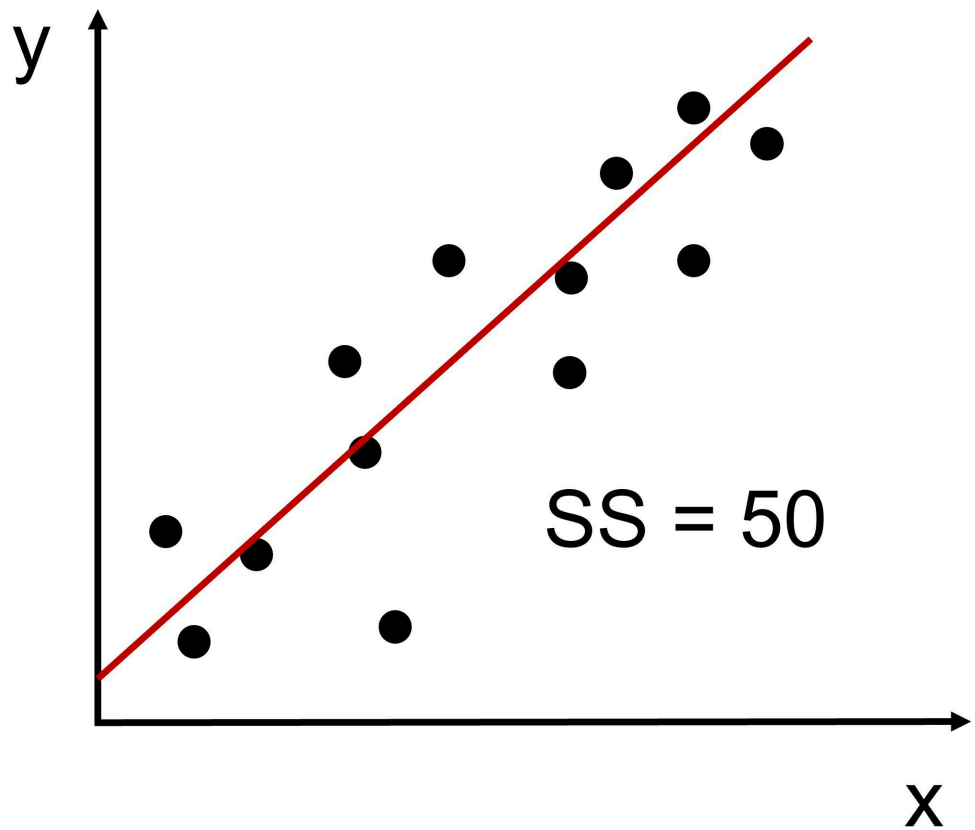


Note

The error is squared so positive and negative errors do not cancel each other out.

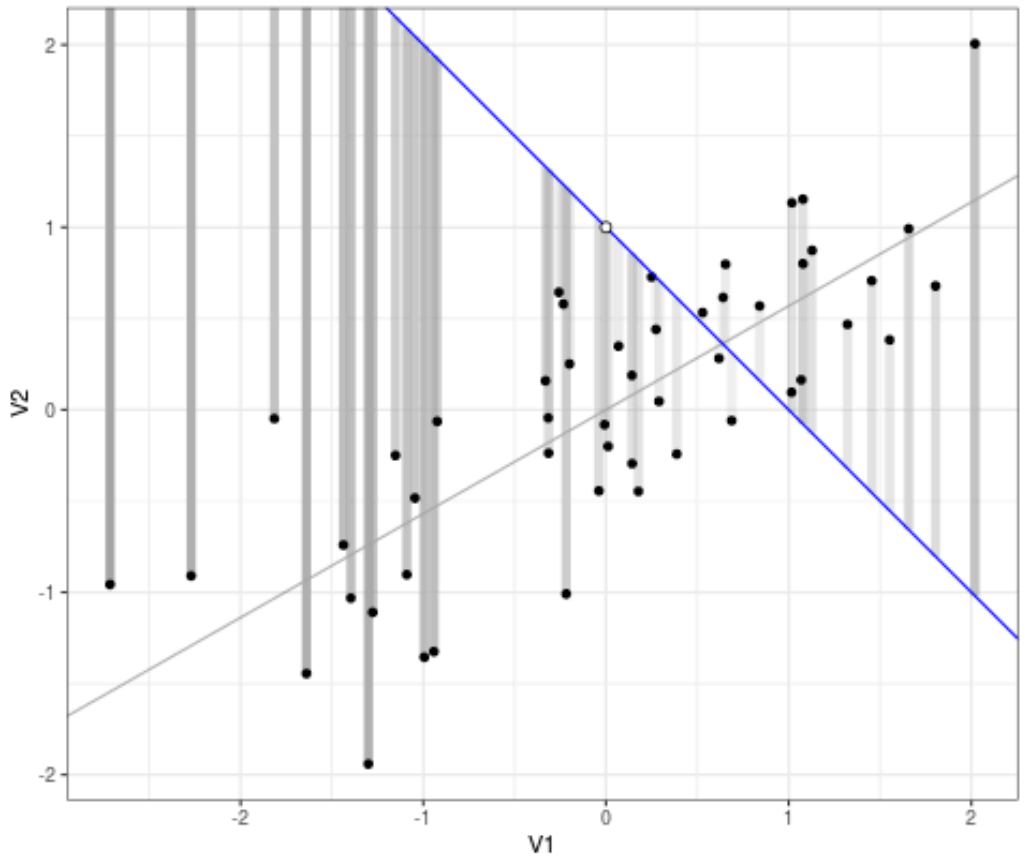
Least squares





Which line fits the data better?

How does a computer fit a line of best fit?


2
1
0
-1
-2
V2
-2
-1
0
1
2
V1

The line is fitted again and again until the squared error stabilises. A computer can do this very quickly!

Simple linear regression

The goal is to fit the line of best fit between a numerical response and a numerical predictor.

Simple linear regression model

We want to relate the response Y to a predictor x for i number of observations:

$$Y_i = \beta_0 + \beta_1 x_i + \epsilon_i$$

where

$$\epsilon_i \sim N(0, \sigma^2)$$

- Y_i, the *response*, is an observed value of the dependent variable.
- β_0, the *constant*, is the y-intercept and is **fixed**.
- β_1 is the population *slope* parameter, and like β_0, is also **fixed**.
- ϵ_i is the error associated with predictions of y_i, and unlike β_0 or β_1, it is *not fixed*.

i Note

ϵ_i is generally associated with the **residual** ($observed - predicted$). **True error** occurs during data collection (e.g. faulty instruments, selection bias, etc.) and often immeasurable.

Different wordings

Different ways to think about the response:

- Response = Prediction + Error
- Response = Signal + Noise
- Response = Model + Unexplained
- Response = Deterministic + Random
- Response = Explainable + Everything else
- Y = f(x)
- Dependent variable = f(Independent variable)

Model fitting

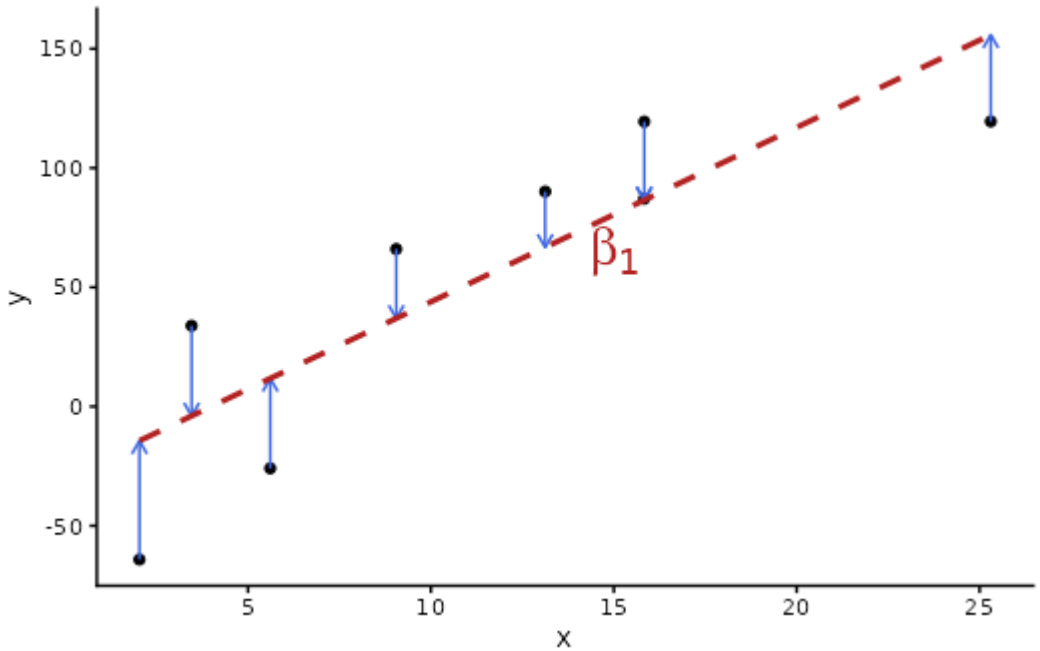
Two approaches; analytical and numerical:

- **Analytical**: equation(s) used directly to find solution
- **Numerical**: computer uses "random guesses" to find set of parameters to that minimises objective function, in this case residual sum of squares

Analytical: Slope, β_1

$$\beta_1 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n}(x_i - \bar{x})^2} = \frac{Cov(x, y)}{Var(x)} = \frac{SS_{xy}}{SS_{xx}}$$

```r
# Calculate slope from df
beta1 <- sum((df$x - mean(df$x)) * (df$y - mean(df$y))) /
  sum((df$x - mean(df$x))^2)
# beta0 <- mean(df$y) - beta1 * mean(df$x)

p1 +
  geom_smooth(method = "lm", se = FALSE, color = "firebrick", linetype = 2) +
  # label the line
  annotate("text",
    x = 15, y = 65, size = 7,
    label = expression(beta[1]), colour = "firebrick"
  ) +
  theme_classic()
```

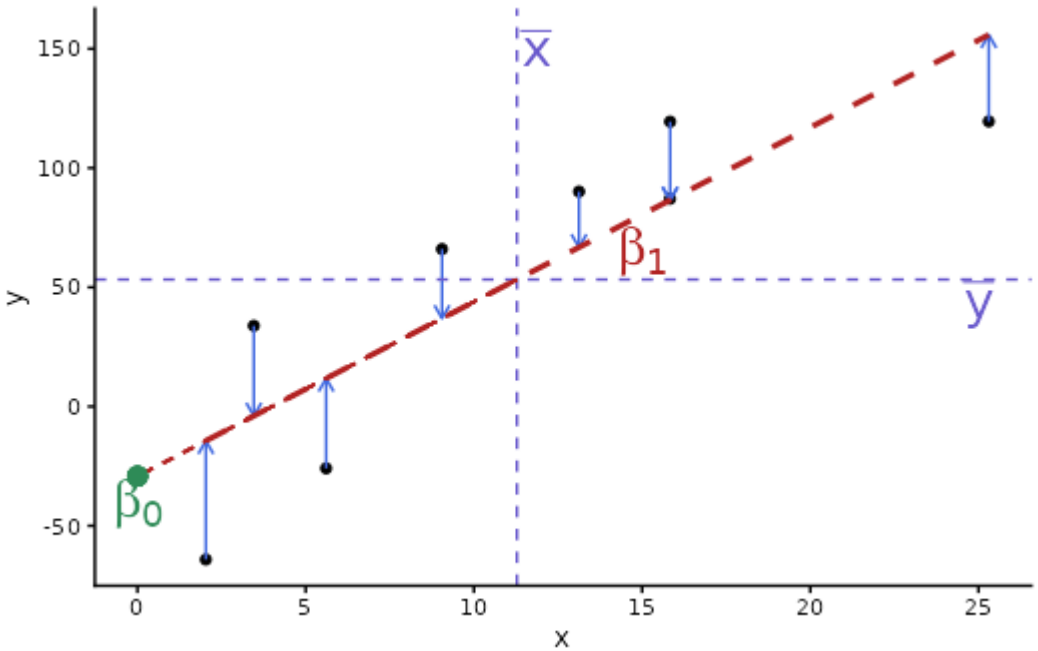

150
100
50
0
-50
y
5
10
15
20
25
x
$\bar{\beta}_1$

Analytical: Intercept

$$\beta_0 = \bar{y} - \beta_1 \bar{x}$$

```r
# calculate mean y from df
ybar <- mean(df$y)
xbar <- mean(df$x)
beta0 <- ybar - beta1 * xbar

p1 + geom_vline(xintercept = xbar, linetype = "dashed", color = "slateblue") +
  geom_hline(yintercept = ybar, linetype = "dashed", color = "slateblue") +
  # label the lines
  annotate("text",
    x = 25, y = ybar * 0.8, size = 7,
    label = expression(bar(y)), colour = "slateblue"
  ) +
  annotate("text",
    x = xbar * 1.05, y = 150, size = 7,
    label = expression(bar(x)), colour = "slateblue"
  ) +
  geom_smooth(method = "lm", se = FALSE, color = "firebrick", linetype = 2) +
```

```r
# extend the geom_smooth line to intercept x=0
geom_segment(aes(x = xbar, y = ybar, xend = 0, yend = beta0),
  color = "firebrick", linetype = 2
) +
# label the slope line
annotate("text",
  x = 15, y = 65, size = 7,
  label = expression(beta[1]), colour = "firebrick"
) +
# add a dot at the intercept
geom_point(aes(x = 0, y = beta0), color = "seagreen", size = 3) +
# label the intercept
annotate("text",
  x = 0, y = beta0 * 1.4, size = 7,
  label = expression(beta[0]), colour = "seagreen"
) +
theme_classic()
```


150
100
50
0
-50
y
0
5
10
15
20
25
x
$\bar{x}$
$\bar{y}$
β_1
β_0

Example: back to Galton's data

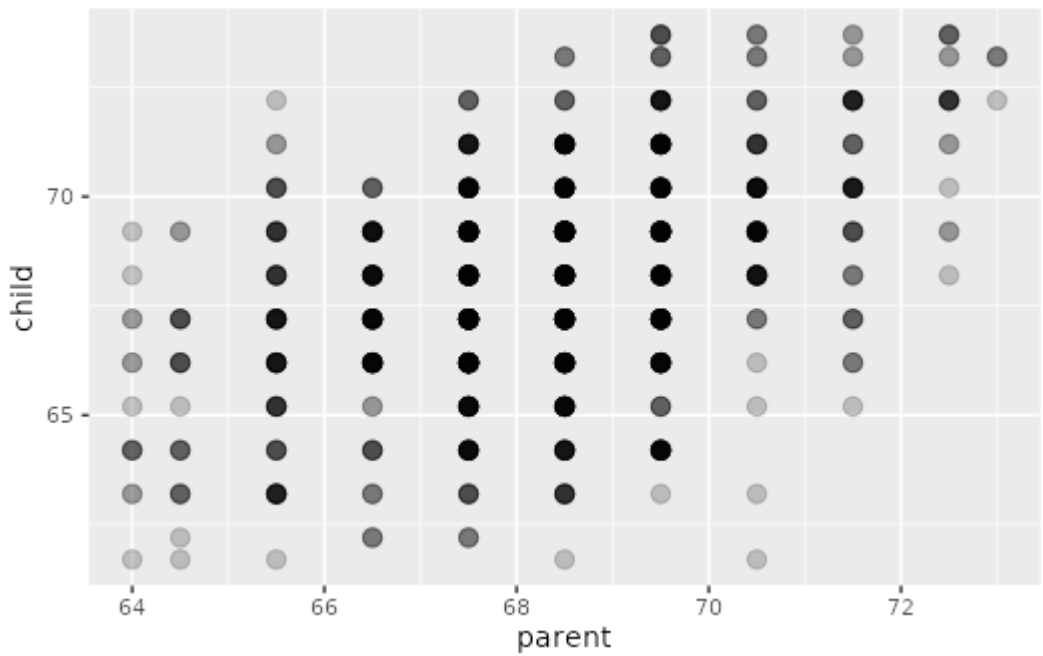
What can we understand about the relationship between *child height* and *parent height*?

> Let's do this in R - much easier

Linearity check

Recap - there appears to be a linear relationship between `child height` and `parent height` (plot). This is a moderately positive relationship (correlation).

```
library(ggplot2)
ggplot(Galton, aes(x = parent, y = child)) +
    geom_point(alpha = .2, size = 3)
```




```r
cor(Galton$parent, Galton$child)
```

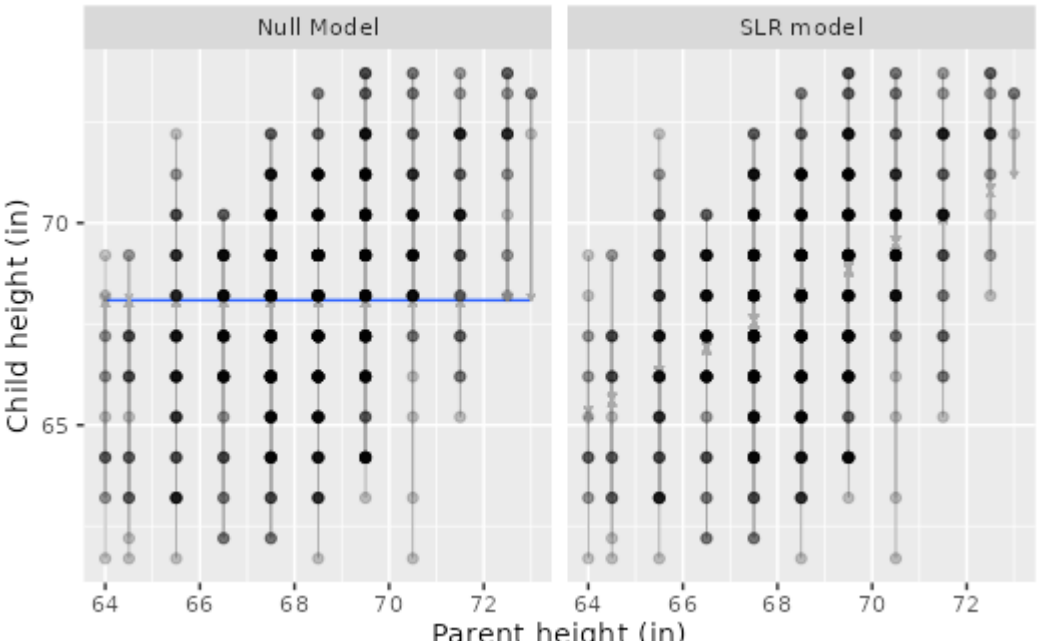
```
[1] 0.4587624
```

Hypothesis testing

- The **null hypothesis** for a linear model: $H_0 : \beta_1 = 0$
  - If there is no slope ($\beta_1 = 0$) then $y = \beta_0$ (the mean)
  - i.e. is the mean ($\bar{y}$) a better fit than a linear model?
- The **alternative hypothesis** for a linear model: $H_0 : \beta_1 \neq 0$
  - i.e. the estimate from the linear model ($\hat{y}$) fits the data better than the mean ($\bar{y}$)

```
null_model <- Galton %>%
  lm(child ~ 1, data = .) %>%
  broom::augment(Galton)
lin_model <- Galton %>%
  lm(child ~ parent, data = .) %>%
  broom::augment(Galton)
models <- bind_rows(null_model, lin_model) %>%
  mutate(model = rep(c("Null Model", "SLR model"), each = nrow(Galton)))

ggplot(data = models, aes(x = parent, y = child)) +
  geom_smooth(
    data = filter(models, model == "Null Model"),
```

```r
    method = "lm", se = FALSE, formula = y ~ 1, size = 0.5
  ) +
  geom_smooth(
    data = filter(models, model == "SLR Model"),
    method = "lm", se = FALSE, formula = y ~ x, size = 0.5
  ) +
  geom_segment(
    aes(xend = parent, yend = .fitted),
    arrow = arrow(length = unit(0.1, "cm")),
    size = 0.3, color = "darkgray"
  ) +
  geom_point(alpha = .2) +
  facet_wrap(~model) +
  xlab("Parent height (in)") +
  ylab("Child height (in)")
```


Null Model
SLR model
Child height (in)
70
65
64
66
68
70
72
Parent height (in)

Simple linear regression

We fit the model in one simple line of code: `fit <- lm(child ~ parent, data = Galton)`

```
fit <- lm(child ~ parent, data = Galton)
```

...

And then we can use `summary()` function to get a summary of the model:

```
summary(fit)
```

```


Call:
lm(formula = child ~ parent, data = Galton)

Residuals:
    Min      1Q  Median      3Q     Max 
-7.8050 -1.3661  0.0487  1.6339  5.9264 

Coefficients:
```

```
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 23.94153    2.81088   8.517   <2e-16 ***
parent       0.64629    0.04114  15.711   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.239 on 926 degrees of freedom
Multiple R-squared:  0.2105,    Adjusted R-squared:  0.2096
F-statistic: 246.8 on 1 and 926 DF,  p-value: < 2.2e-16
```

Summary

Correlation is a measure of the relationship between two numerical variables between -1 and 1.

- A correlation coefficient measures the *strength* and *direction* of the relationship.
- Pearson's correlation is for linear relationships, Spearman's/Kendall's correlation is for monotonic relationships.
- Correlation $\neq$ causation.

Regression is an analysis that models the relationship between a dependent variable and independent variable(s).

- The most common method for regression is least squares, which minimises the sum of the squared residuals.
- Simple linear regression fits a straight line between two variables.

> Next week: interpreting results and assumptions for simple linear regression.

Thanks!

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